

Problem to be modeled

- Barista serving costumers
- This is a single person coffee stall so one server
- Jobs will arrive at a variable rate
- Some customers will have multiple items. We will only do item and put the remaining orders into our “sink”

How do we map this?

- We will assume for our purposes all coffee tasks are the same operation
- It will take 10 timesteps to fulfill the task of serving coffee
- As this is a single stall we have one queue with one server attached to it

What are the metrics that matter?

- How fast a customer gets their product once in the queue is the most important metric
- Wait
- Response time
- Latency
- Utilization
- Errors

How do you map that queueing problem in terms of the simulator?

- As this is a single queue problem we can manually generate our queue configuration
- Our workload will be generated from a normal distribution
- We will graph the previously noted important metrics.

Run the simulation at various scales

- When running the simulation it is apparent that it is easy for a single server to get backed up
- This was expected as we are simulating an overworked single employee coffee stand

Sensitivity Study

- We see that the main issue we run into is a massive backup of jobs waiting for our one server to be done
- To simulate another server being added that would work together to half the work time, I will make the server go twice as fast.
- After doing this we can see that the time spent waiting is much less.
- If you wanted to destroy response time you could do that by turning any of the knobs that affect how fast our server is able to make “coffee” it is already brutal at 10 timesteps as can be seen in the supporting figures.

Takeaways

- It is very inefficient for one person to serve so many customers(jobs) in this sort of arrangement.
- Work can pile up quickly when we are receiving at a rate much higher than we can service.
- The bottleneck here is the server's ability to do work.
- Improvements such as adding more queue space or another queue would not help as there is just one server.